National University
Of Computer and Emerging Sciences

Project Phase #2

Presented to

Dr. Basharat Hussain

In partial fulfillment

of the requirements for the course of

AI Product Development (AI - 4013)

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MealMind: Reducing Daily Meal Decision Fatigue

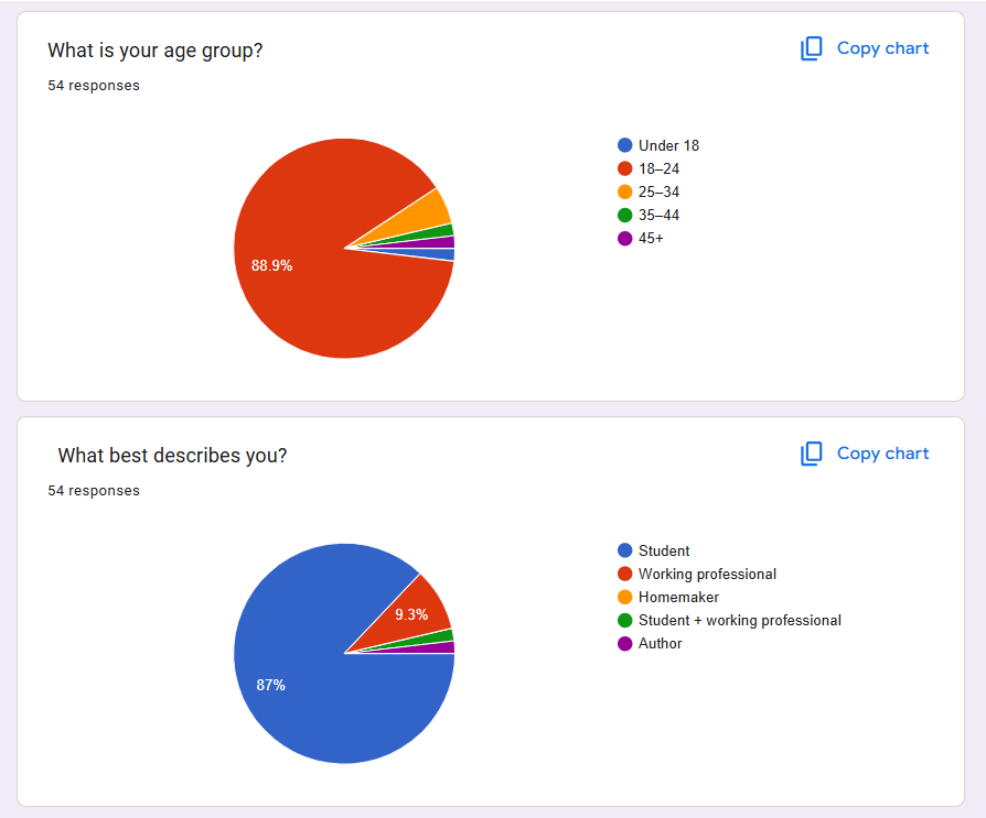
Phase 2: Research Report; Product, Business & Market Evidence

1. Product Research

A. User Needs & Pain Points

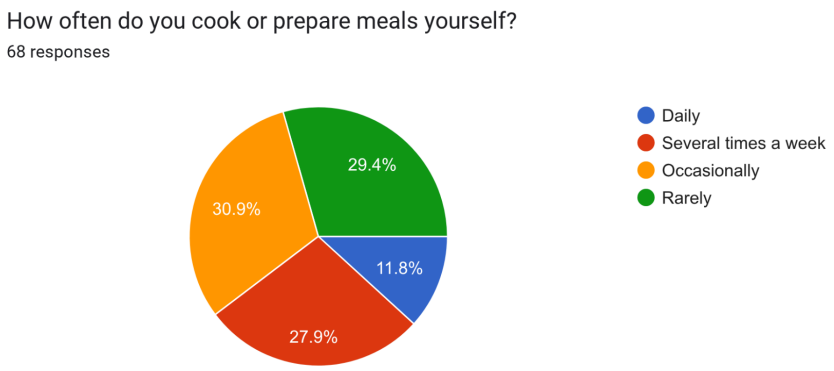
Primary research was conducted via a structured Google Form survey of 54 respondents. The survey covered the following questions:

Q1. What is your age group? Q2. What best describes you?



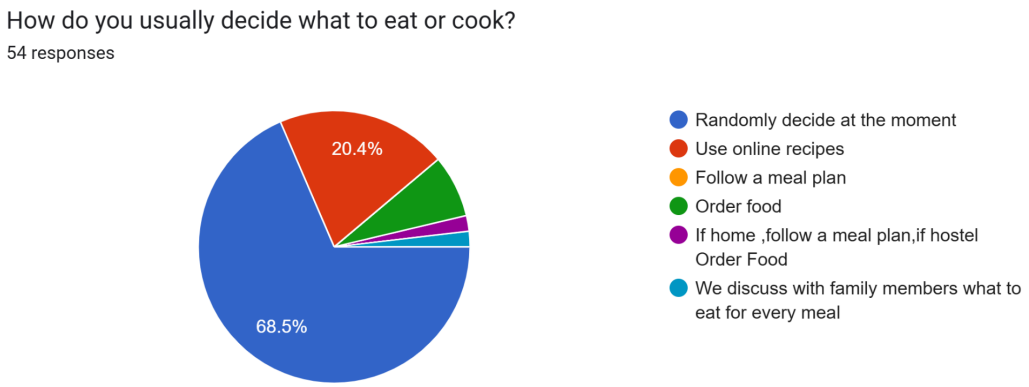
The overwhelming majority (89%) fell in the 18–24 age bracket, with students comprising 86% of respondents. A smaller portion were working professionals aged 25–44. This strongly confirms that university students and young professionals are the accessible early adopter segment for MealMind.

Q3. How often do you cook or prepare meals yourself?



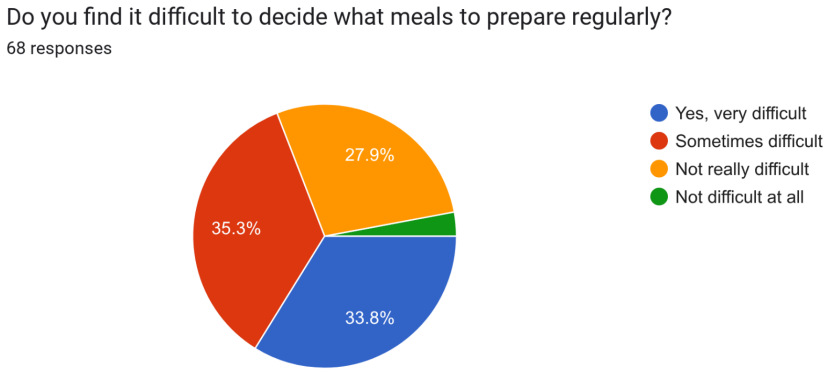
Respondents cook with varied frequency — many cook occasionally or several times a week but rely heavily on spontaneous decisions rather than any structured planning system.

Q4. How do you usually decide what to eat or cook?



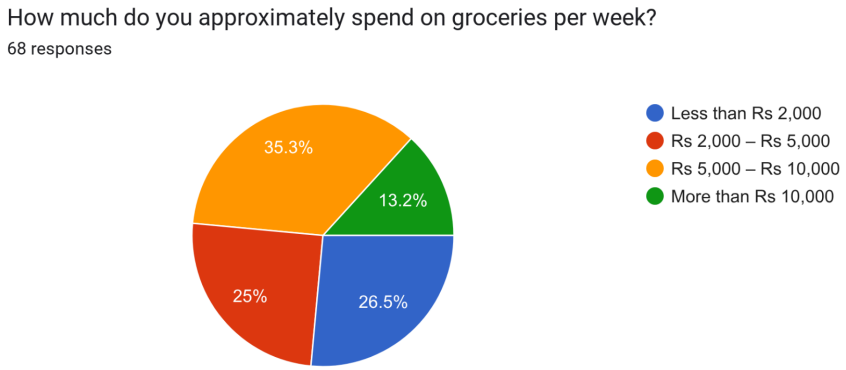
The dominant pattern was "randomly decided at the moment," cited by the majority of respondents. This directly validates the existence of unstructured, high-effort decision-making at mealtime. A smaller group uses online recipes, and very few follow any structured meal plan.

Q5. Do you find it difficult to decide what meals to prepare regularly?



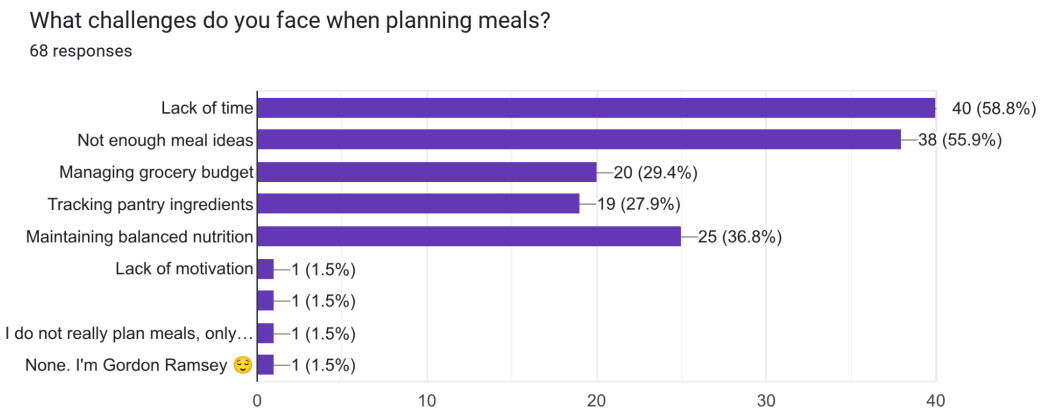
39% of respondents said meal decisions are "Yes, very difficult," and a further 32% said "Sometimes difficult." Only 18% reported no real difficulty. This is the core validation of MealMind's problem statement — decision fatigue is real, frequent, and widely felt within this demographic.

Q6. How much do you approximately spend on groceries per week?



Grocery spending was concentrated in the PKR 2,000–PKR 10,000 per week range, with a notable portion spending below PKR 2,000. This confirms the price-sensitive nature of the target segment and validates the importance of budget-aware meal suggestions as a core product feature.

Q7. What challenges do you face when planning meals?

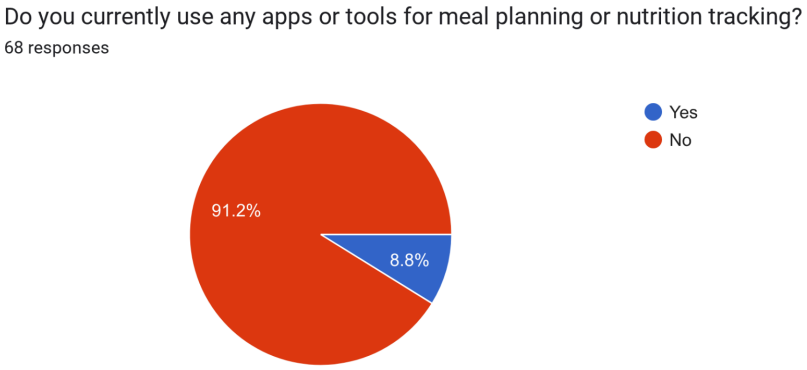


Respondents could select multiple challenges. The most frequently cited were:

- Not enough meal ideas / variety — ~64% of respondents
- Lack of time — ~60% of respondents
- Maintaining balanced nutrition — ~57% of respondents
- Tracking pantry ingredients — ~46% of respondents
- Managing grocery budget — ~43% of respondents

B. Existing Solutions & Substitutes

Q8. Do you currently use any apps or tools for meal planning or nutrition tracking? If yes, which apps do you use?



86% of survey respondents reported using no dedicated meal planning app. Of those who use any tool at all, responses included Foodpanda (for food delivery, not planning), ChatGPT (used informally for meal suggestions), and recipe websites. This near-zero adoption of dedicated

tools is a strong signal — not that the market doesn't want a solution, but that existing solutions haven't solved the problem for this demographic.

Q9. What do you dislike about existing meal planning apps or tools?

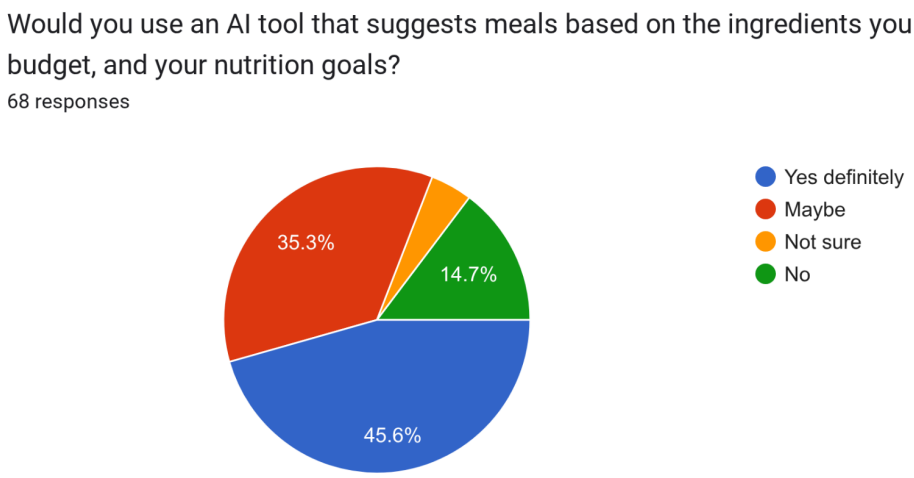
Key themes from open-ended responses included:

- All paid options are expensive; free options are dysfunctional
- No tools are designed around the Pakistani diet or local ingredients
- Foodpanda provides suggestions but doesn't help with nutrition or budget decisions
- Apps are too generic and not context-aware
- One respondent used ChatGPT as a workaround but noted it is expensive for premium access

Indirect substitutes currently in use include Google/recipe websites, random in-the-moment decisions, family consultation, and ChatGPT. These are zero-cost and low-effort, which means MealMind must deliver a meaningfully better experience — not just "also free."

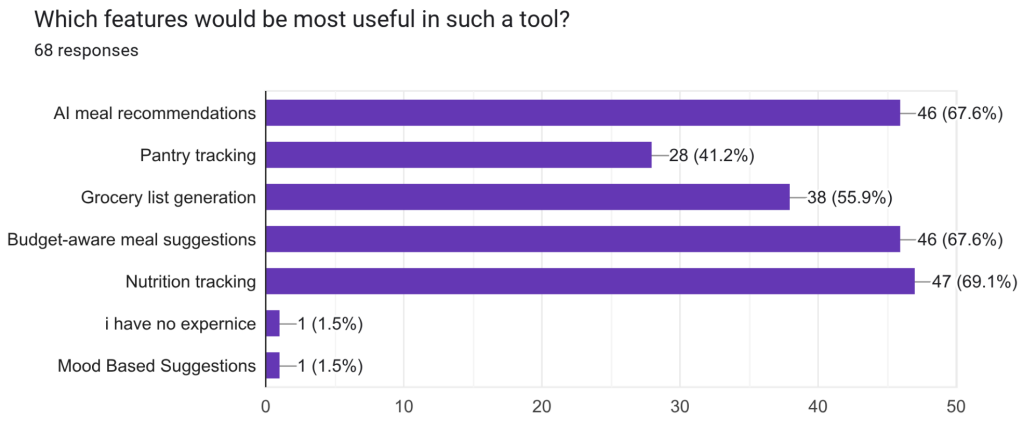
C. Product Scope & Differentiation

Q10. Would you use an AI tool that suggests meals based on the ingredients you already have, your budget, and your nutrition goals?



Response was strongly positive. The majority said "Yes, definitely," with very few saying no. This is direct validation that the core MealMind concept — contextual AI-driven meal suggestions — resonates with the target segment.

Q11. Which features would be most useful in such a tool?



Feature preferences ranked by selection frequency:

- AI meal recommendations — ~70% of respondents
- Budget-aware meal suggestions — ~70% of respondents
- Nutrition tracking — ~68% of respondents
- Grocery list generation — ~61% of respondents
- Pantry tracking — ~44% of respondents

The core product must suggest contextually relevant meals based on what the user has, what they can afford, and what their nutritional needs are — at the moment the decision needs to be made. Pakistani cuisine and local ingredient context is a non-negotiable differentiator that emerged directly from open-ended responses.

What Will MealMind NOT Attempt to Solve?

Based on Phase 1 critique and this research, MealMind will explicitly not pursue receipt scanning, store price comparison, expiry tracking, or enterprise/grocery retailer partnerships in the MVP. These features increase scope and complexity beyond what is validated.

Differentiation

MealMind's differentiation rests on three evidence-backed pillars: (1) Pakistani diet and local food context — no competitor addresses this; (2) integrated decision-making — not just recipes but budget + nutrition + pantry in one moment; (3) reducing cognitive load rather than adding logging burden. Existing apps add effort (MyFitnessPal), are expensive (HelloFresh), or are too generic (Eat This Much). MealMind's value is in reducing friction at the decision point.

2. Business Research

A. Customer & Buyer Dynamics

At this stage, the user and buyer are the same person. Survey respondents are students and young professionals who make their own food purchasing decisions. Budget spend is concentrated in the PKR 2,000–PKR 10,000/week grocery range, with many below PKR 5,000. This confirms price sensitivity as a central business constraint.

There is no validated B2B buyer (e.g. grocery retailers, health brands) at this stage. Any such pathway is speculative and should be deferred post-validation. Purchasing decisions are individual and convenience-driven — users will adopt if onboarding is effortless and value is immediate.

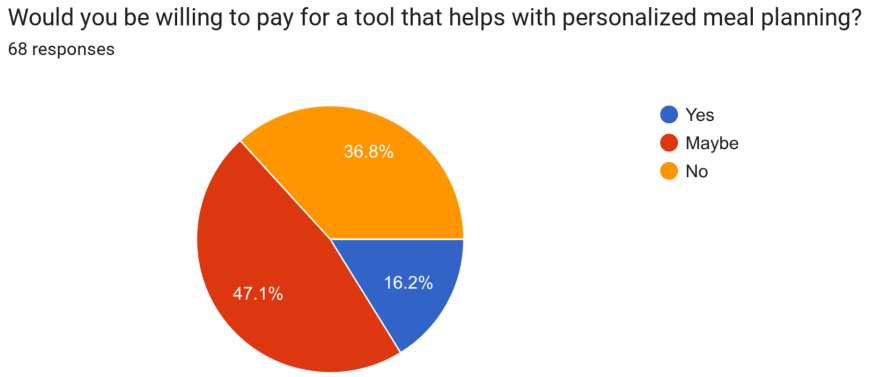
B. Revenue & Cost Logic

Pricing Benchmarks

Comparable apps in the global market: MyFitnessPal Premium (~PKR 5,500/month), Mealime Pro (~PKR 1,650/month), Cronometer Gold (~PKR 2,750/month). In the Pakistani context, digital subscription willingness is significantly lower than global benchmarks, and PKR 200–500/month is a realistic ceiling for a student segment.

Willingness to Pay — Survey Evidence

> *Q12. Would you be willing to pay for a tool that helps with personalized meal planning?*



Of 54 respondents:

- Yes — 11 respondents (20%)
- Maybe — 25 respondents (46%)
- No — 18 respondents (33%)

This result shows moderate willingness to pay among respondents. About 20% of respondents said they would be willing to pay, while 46% selected “maybe,” indicating that many users are open to paying if the product demonstrates clear value. However, 33% stated they would not pay, which highlights the price sensitivity of the target segment, particularly students.

Overall, the results suggest that while direct willingness to pay is limited, there is still meaningful potential for monetization if the product delivers clear benefits. Based on these findings, a freemium model appears to be the most suitable approach, where core features remain free and advanced features are offered through a low-cost subscription.

Primary Cost Drivers

Key costs will include LLM/AI API usage per recommendation call, cloud hosting and compute, frontend and backend development, and ongoing model prompt tuning. The risk — flagged in Phase 1 — is that AI costs per active user may outpace subscription revenue if the free tier is heavily used. This must be managed through query throttling on free accounts.

C. Business Risks & Constraints

- **Low conversion:** Price-sensitive users may not upgrade from the free tier — mitigated by limiting advanced features (e.g. nutrition analytics, budget tracking history) to the paid tier

- **Trust in AI outputs:** Incorrect nutritional information could damage credibility — requires clear disclaimers and validated nutritional data sources

- **Data privacy:** Dietary and health data is sensitive — requires compliant storage and a transparent data use policy

- **Feature creep:** Phase 1 revealed scope risks; the MVP must be ruthlessly narrowed

- **Competitor response:** ChatGPT and large AI tools could be prompted for similar outputs at no cost — MealMind's advantage must be UX convenience and persistent personalisation, not AI capability alone

3. Market & Industry Research

A. Market Context

Target Market Definition

Primary: Pakistani university students and young professionals (ages 18–30) who cook for themselves, manage their own grocery budgets, and have some interest in eating healthily. Secondary: busy households managing family meals on constrained budgets.

Market Size — Proxy Indicators

Pakistan's app economy is growing rapidly. Mobile internet penetration exceeded 50% of the population as of 2024, with smartphone adoption driven heavily by the 18–34 age group. Pakistan's university student population exceeds 1.7 million. The health and fitness app market in South Asia has seen consistent double-digit growth year over year. While a Pakistan-specific meal planning app market size is not publicly documented, the behavioral indicators — high cooking frequency, grocery price sensitivity due to inflation, and near-zero app adoption — indicate an underserved market with strong latent demand.

Relevant Industry Trends

- Food inflation in Pakistan has made home cooking more financially urgent, increasing demand for budget optimization tools
- AI-powered consumer apps are gaining mainstream acceptance among 18–30 demographics
- Growing interest in nutrition and fitness among urban Pakistani youth
- Expansion of local grocery delivery platforms signals growing digital food behaviour in Pakistan

B. Competitive Landscape

Competitor	Type	Strength	Gap
Eat This Much	Direct	Auto-generates calorie-based meal plans	No pantry awareness, no budget, no local food context

Mealime	Direct	Grocery list generation from recipes	Manual meal selection, no nutritional optimization, paid
MyFitnessPal	Direct	Large food database, strong nutrition logging	Reactive (logs after eating), no planning or budget, expensive premium
ChatGPT	Indirect	Free, flexible, context-aware if prompted well	No persistent user profile, no pantry/budget memory, high effort per session
Foodpanda	Indirect	Familiar to Pakistani users, large restaurant selection	Expensive, delivery-only, no health/budget awareness, causes decision overload
Pinterest / Recipe Sites	Substitute	Free, visually engaging, large content library	No personalization, no budget/nutrition awareness, browsing is itself a form of decision fatigue

No single tool wins comprehensively. Users patch together ChatGPT, Pinterest, and random decisions because no integrated, culturally-relevant tool exists for Pakistan. The gap MealMind fills is real and unoccupied.

C. Structural Barriers

- **Switching costs are low** — users can abandon MealMind easily since data portability is not yet a lock-in factor. The MVP must build habits quickly through daily value delivery.
- **Cultural resistance** — Pakistani users may be skeptical of AI nutrition advice. Framing around practical convenience (saving money, saving time) rather than "health optimization" may lower resistance.
- **No significant regulatory barriers** exist for a meal suggestion app in Pakistan at this stage, though health-claim regulations should be monitored.
- **Language and localization** — an English-only app limits reach. Future Urdu interface support would be a meaningful barrier-reducer.

4. Economic Research — Macro Risks

Pakistan's Macro Context

Pakistan has experienced severe food inflation over 2023–2025, with grocery prices rising significantly across staple categories. This macro condition is actually a tailwind for MealMind: when grocery budgets are under pressure, a tool that helps users cook smarter with what they have becomes more valuable, not less. Budget-aware meal suggestions and waste reduction are especially relevant in this environment.

Key Macro Risks:

- **Inflation & purchasing power:** While food inflation supports demand for budget tools, it also means users have less disposable income for app subscriptions. The freemium model must be generous enough to deliver real value in the free tier.
- **Political and economic instability:** Prolonged economic uncertainty in Pakistan may suppress discretionary digital spending. Subscription pricing must remain very low — PKR 200–500/month is a realistic ceiling.
- **Internet and smartphone penetration gaps:** While urban penetration is high, rural and semi-urban access remains uneven. MealMind's early market is urban-university-centric, which mitigates this risk for the MVP.
- **Regulatory risk:** AI products globally are facing increasing scrutiny. While Pakistan has no imminent AI regulation, nutrition-related AI advice may eventually face health claim restrictions. Disclaimers and non-prescriptive language should be built into the product from day one.
- **Currency and payment infrastructure:** Many Pakistani users lack credit cards or international payment methods. Local payment gateway integration (JazzCash, Easypaisa) is essential for any subscription revenue.
- **Data localisation:** If Pakistan moves toward data sovereignty regulations, local hosting of user dietary data may become a legal requirement. Architecture should plan for this from the outset.

5. Assumptions & Evidence Tracker

Assumption	Status	Evidence / Notes
Users experience significant daily meal decision fatigue	Confirmed	71% of respondents reported decisions as "very difficult" or "sometimes difficult." High-frequency problem confirmed.
Students and young professionals are the primary early adopter segment	Confirmed	86% of respondents are students; 89% aged 18–24. Segment is accessible and behaviourally active.
No existing tool serves the Pakistani diet and local food context	Confirmed	Multiple open-ended responses explicitly cited the absence of Pakistan-specific apps as a core pain point.
Users are willing to pay for a personalized meal planning tool	Partially Confirmed	20% said "yes," 46% said "maybe." Freemium is the right model; hard conversion requires demonstrated value first.
AI meal recommendations are the most valued feature	Confirmed	75% selected AI meal recommendations as their top desired feature in the survey.
Integration of budget + nutrition + pantry delivers more value than any single tool	High Risk	Assumed but not directly tested. Users may find value in simpler tools. MVP must test this.
Users will adopt a new app rather than continue using ChatGPT or recipe sites	High Risk	86% use no app today. Convenience of informal alternatives is high. Switching cost to MealMind must be near zero.
The product can be built at a cost sustainable relative to early revenue	High Risk	AI API costs scale with usage. With low or PKR 0 revenue in the early phase, cost management is critical. Unresolved.
Busy parents and families are a viable secondary segment	High Risk	Only 1 respondent was 45+. Survey did not validate this segment. Deprioritise for MVP.
Enterprise/B2B (grocery retailers) is a viable revenue channel	High Risk	Entirely unvalidated. No evidence in research. Remove from MVP scope entirely.

6. Phase 2 Summary

Research confirmed that meal decision fatigue is a real and validated problem for Pakistani students and young professionals, with the absence of locally-relevant tools standing out as MealMind's strongest differentiator. Key monetisation and integration assumptions, however, remain unresolved and will be tested in Phase 3.

What Has Changed?

Phase 1 proposed a broad feature set — receipt scanning, expiry tracking, store price comparison, and enterprise partnerships. None of these were validated by this research. All have been cut. MealMind's focus is now singular: reduce the effort of deciding what to cook, using ingredients the user already has, within a real budget, grounded in Pakistani food culture.

What Will the MVP Focus On?

The Phase 3 MVP will test one hypothesis: that a contextual AI meal suggestion tool built around Pakistani cuisine delivers enough immediate value to form a daily habit. It will include:

- AI-generated meal suggestions based on pantry inputs, weekly PKR budget, and nutrition preferences
- Pakistani cuisine as the primary recipe and ingredient database
- Grocery list generation and basic per-meal nutrition information
- Freemium model with a target paid tier of PKR 200–500/month

Explicitly excluded until core value is proven: receipt scanning, price comparison, expiry tracking, mood-based suggestions, and all enterprise features.